

Resolution Geometry

Maximal Closure: Typed Statuses, Composite Soundness and Stationary Slices

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October 2026

Abstract

Every admitted RG request ends in exactly one of seven statuses: LOCKED, ESCALATE, CORRESPONDENCE, VACUOUS, INCONSISTENT, STRUCTURAL WALL[ε] or OUTSIDE DOMAIN. It is LOCKED iff every gate of the requested module is certified. Totality and determinism are structural properties of the dispatcher. Its soundness is inherited from its certificates: $\Pr(\text{false LOCKED}) \leq \sum_g \alpha_g$. The dispatcher was wired to the real rank, Petz-band and anchor certificates. On 2000 instances with known truth (five seeds) it gave 0 false LOCKED and power 1.0, while three sabotaged certificates gave false LOCKs. Walls are window-indexed. The stationary-slice extension is locked: radar distance is the Landau–Lifshitz metric, and loop synchronization carries Sagnac holonomy. Battery: 6/6 on five seeds, 3/3 canaries.

1 Results

Definition 1 (Statuses). LOCKED, ESCALATE, CORRESPONDENCE, VACUOUS, INCONSISTENT, STRUCTURAL WALL[ε], OUTSIDE DOMAIN, with gates evaluated in the order: domain, consistency, task, vacuity, rank, candidate, scale.

Theorem 2 (Totality and determinism; CL01). *On all 384 combinations of certificate outcomes, exactly one status is returned, and LOCKED iff all gates pass. All seven statuses are reachable.*

This is a property of a decision procedure. Exhaustive enumeration of Boolean inputs therefore checks the code, not RG.

Theorem 3 (Composite soundness; CL02, CL03). $\Pr(\text{false LOCKED}) \leq \sum_g \alpha_g$ (union bound).

- *With the double-escalation rank certificate, the nonparametric Petz band and the declared anchor, the five seeds gave 132–179 locks each and 0 false locks, with power 1.0.*
- *Under the same dispatcher, numerical rank of noisy data, a panel argmin and a silently assigned scale gave 60–84, 16–32 and 8–14 false locks per seed.*

Proposition 4 (Window-indexed walls; CL04). *A true wall ($\sigma_{\min} = 0$) and a full-rank law with $\sigma_{\min} = 0.003 < \tau = 0.005$ both end as STRUCTURAL WALL[τ], while $\sigma_{\min} = 0.08$ is LOCKED.*

Proposition 5 (Stationary slices; CL05, CL06). *For observers co-rotating at $\Omega R = 0.5$ ($c = 1$):*

- *radar distance converges to the Landau–Lifshitz distance with relative errors 9.9×10^{-3} , 2.5×10^{-3} , 6.3×10^{-4} and 1.6×10^{-4} (order 1.99), while the inertial arc length is off by 15%;*

- *the Sagnac proper-time difference equals $4\Omega A/\sqrt{1-\Omega^2 R^2}$ to machine precision, and it vanishes iff $\Omega = 0$.*

No-Go 6 (Audit of the earlier combined dispatcher battery). Its inputs were hand-set booleans. Its “independent oracle” was a hand-written table for the named cases, which its stress test then sampled. Its gate cube re-read the dispatcher’s own branches. Its canaries were `lambda x: 'LOCKED'`. It also issued STRUCTURAL WALL from a boolean, which L38 forbids for data.

Its decision order and status set are retained here, wired to computed certificates.

Reproducibility

`rg_closure_attack.py` passes 6/6 on seeds 20261003, 1, 42, 777 and 2026. Canaries (argmin, numerical rank, silent scale) are all caught. Correction before registration: the first rank sabotage compared a point estimate with τ . It was conservative at this noise level and produced no false locks, so it was replaced by numerical rank.